Customer Churn Analysis Report – Zelcom

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Data Period: 2019–2024
Objective: Understand and reduce customer churn through data-driven insights.

Problem Statement

Problem statement: The Churn Challenge
18% of customers churn within the first 6 months
Lack of visibility into when, why, and who is churning
No targeted retention strategies

Objectives:
Understand churn trends
Identify churn drivers
Segment customers by churn risk
Monitor retention performance

Data Cleaning and Preparation (MySQL)

Before analysis and visualization in Power BI, the raw customer dataset underwent extensive data cleaning in MySQL to ensure accuracy and reliability of insights. The following key steps were performed:

Data Validation & Integrity Checks

- Verified that CustomerID was unique and served as the primary key.
- Checked for null or duplicate values across key fields (e.g., customerID, tenure, MonthlyCharges, TotalCharges).

Handling Missing Values

- Identified blank/NULL entries in TotalCharges.
- Replaced missing values with 0 where tenure = 0 (new customers).
- Removed rows with incomplete critical data (where cleaning wasn't possible).

Data Type Corrections

- Converted TotalCharges from string (VARCHAR) to numeric (DECIMAL) for accurate calculations.
- Ensured date fields (e.g., CustomerSince) were in DATE format for proper filtering and time-based analysis.

Standardization

- Normalized categorical variables (e.g., ensuring "Yes/No" instead of mixed entries like "Y/N", "Yes/No").
- Standardized gender to "Male" and "Female" only.

Derived Columns for Analysis

- Created a Churn Flag (1 = churned, 0 = active) for modeling.
- Calculated Average Monthly Revenue per Customer using:
- SELECT CustomerID, (TotalCharges / NULLIF(tenure,0)) AS AvgMonthlyRevenue
- FROM Customers;
- Extracted Year & Month from CustomerSince for time-based churn analysis.

Outlier Detection

- Flagged unusually high/low values in MonthlyCharges and TotalCharges using interquartile ranges.
- Investigated and corrected data entry errors (e.g., negative charges).

Final Clean Dataset

- Exported the clean table cleaned_data for loading into Power BI.
- Ensured referential integrity between customer data, services data, and billing records.

Insights

Objective 1: Understand Churn Trends Over Time

Key Findings:

- **Overall churn rate** over the last 6 years is **27%**, with a notable concentration in **November**.

- **Average churn tenure** is **17 months**, indicating customers often leave after more than a year of service.
- Customers with monthly charges above $80 churned twice as often as lower-paying customers.
- New customers (<6 months) were the most likely to churn, while long-term clients (>24 months) had churn rates below 10%.
- A spike in November churn (35%) showed that short-term deals attract customers but fail to retain them.

- There's a **churn spike at two ends**:
  New customers (low tenure) and long-term users (high tenure), suggesting onboarding gaps and retention fatigue.

Objective 2: Identify Key Drivers of Churn

Key Insights:

- **Demographics at higher risk: Male senior citizens**.
  Male senior citizens show the highest churn at 50.26%.
  Customers with month-to-month contracts are most vulnerable. These customers are the majority with churn rate at 43%.

- **Contract length matters**: Customers on **month-to-month contracts** churn more than long-term contract holders.

- **Internet service plays a role**: Customers **without internet service** show significantly higher churn.
  Fibre Optic customers have the highest number of churning rates at 42%.
  Customers with no internet service are more likely to churn.
  Usage patterns show low engagement → higher churn risk.

- **Billing relationship:** Higher monthly charges **may contribute** to churn, though further analysis is recommended.

- No strong evidence ties churn to any particular **subscription type**.

Objective 3: Segment Customers Based on Churn Risk

Risk Profile:

- **High-risk characteristics:** Month-to-month contracts, short tenure, low charges.

- **Churn risk scoring** was derived by segmenting customers based on their **internet service type** (Fibre, DSL, No Internet).

- **High-risk segments:** Primarily **low-tenure** and **high-charge** customers; **high-tenure churners** may represent edge cases.

- **Customer distribution:** Majority of customers fall under the **low churn risk** category.
- **Customer Lifetime Value (CLV):** Highest for two-year contracts (~4K), lowest for month-to-month (~1.5K).

- **Risk Score:** Majority of retained customers have the lowest churn risk score.

- **Revenue Contribution:** Two-year contract customers generate the most revenue (~6M).
- **Senior Citizens & Payment:** Most senior citizen customers use electronic checks, a group that shows high churn.

Next Steps

1. **Focus retention efforts** on early-tenure customers through onboarding and engagement.
2. **Promote long-term contracts** to reduce risk of churn.
3. **Investigate churn among high-tenure users** to identify satisfaction drop-off.
4. **Offer internet-based bundles** to increase perceived value and reduce churn for no-internet customers.
5. **Refine churn scoring models** to enable targeted interventions and A/B testing.